

AWS STUDENT COMMUNITY DAY · ISLAMABAD

FROM CODE TO CLOUD

Your first DevOps pipeline on AWS

FarhanAshrafDev · Live session

Quick show of hands

01

Who has built an application?

02

Who has deployed one to the internet?

03

Who has accidentally broken production?

Today, everyone gets from the first question to the second—safely.

CLOUD IN 30 SECONDS

Rent computing. Use it on demand.

0 servers to buy before your first deployment

Elastic

Grow or shrink when demand changes.

Programmable

Create infrastructure through APIs and code.

THE REAL PROBLEM

“It works on my machine.”

How does every change reach users **quickly**, **safely**, and **repeatedly**?

Manual deployments depend on memory. Reliable deployments depend on a system.

DEVOPS IS NOT A JOB TITLE

DevOps is a way of working

Culture

Developers and operations own outcomes together.

Practices

Small changes, feedback, measurement, learning.

Automation

Machines repeat the boring, risky steps.

Before vs. with DevOps

Manual

- Big releases
- Hero debugging
- Fear of Friday
- Slow feedback

Automated

- Small releases
- Repeatable checks
- Easy rollback
- Fast feedback

Speed comes from **safety**.

CI/CD AS A GROUP PROJECT

Integrate early. Deliver often.

CI · Combine work frequently

NEXT

Build and check every change

NEXT

CD · Keep it ready to release

NEXT

Deploy with a repeatable process

THE DELIVERY LOOP

One change. Five stages.

1 • CODE

[NEXT](#)

2 • BUILD

[NEXT](#)

3 • CHECK

[NEXT](#)

4 • DEPLOY

[NEXT](#)

5 • OBSERVE

ONE POSSIBLE AWS TOOLBOX

Match tools to stages

Source

GitHub · CodeCommit

Build and delivery

GitHub Actions · CodeBuild · CodePipeline

Run

S3 · Lambda · ECS · EKS · EC2

Observe

CloudWatch · X-Ray · CloudTrail

The practice matters more than the product list.

TODAY'S ARCHITECTURE

Git push to live website

GITHUB REPOSITORY

TO

GITHUB ACTIONS

TEMPORARY IDENTITY VIA OIDC

PRIVATE AMAZON S3

SECURE CONTENT DELIVERY

AMAZON CLOUDFRONT

DEMO MISSION

Change one line. Ship it live.

- ☒ Edit the release message
- ☒ Commit and push
- ☒ Watch automated deployment
- ☒ Refresh the live website
- ☒ Prove no AWS key was stored

Audience prediction: will the pipeline go green?

NO SLIDES FOR THE NEXT 8 MINUTES

LIVE DEMO

Code / Commit / Pipeline / AWS

WHAT JUST HAPPENED?

The computer did the checklist

GitHub detected the commit

[NEXT](#)

The workflow checked its inputs

[NEXT](#)

OIDC issued temporary access

[NEXT](#)

Files synced to private S3

[NEXT](#)

CloudFront refreshed the website

WHAT IF A CHECK FAILS?

Stop the conveyor belt.

A failed pipeline is **good news early**, not bad news in production.

Fix forward

Correct the change and redeploy.

Roll back

Restore the last known-good version.

DEVSECOPS

Security travels with the code

No permanent AWS keys

OIDC creates short-lived credentials.

Least privilege

The role can update only this site.

Trace every change

Git history and AWS logs preserve evidence.

DEPLOYMENT IS NOT THE FINISH LINE

Operate the whole lifecycle

Infrastructure as Code

Recreate environments reliably.

Observability

Know what users experience.

Rollback

Recover quickly from failure.

Feedback

Improve the next small change.

YOUR FIRST 30 DAYS

Learn by shipping

WEEK 1 · Git + Linux basics

[NEXT](#)

WEEK 2 · AWS IAM, S3, CloudFront

[NEXT](#)

WEEK 3 · CI/CD + containers

[NEXT](#)

WEEK 4 · IaC + monitoring

Build one tiny project. Automate it. Break it. Recover it. Explain it.

REMEMBER ONLY THREE THINGS

Small. Automated. Observable.

01 · Small changes

Reduce risk and simplify recovery.

02 · Automated path

Make delivery repeatable.

03 · Fast feedback

Learn before users suffer.

SLIDES · CODE · AWS TEMPLATE · RUNBOOK

Thank you. Questions?



**[github.com/FarhanAshrafDev/
devops-on-aws-community-day](https://github.com/FarhanAshrafDev/devops-on-aws-community-day)**

Source references and cleanup instructions are in the repository.